Core-flooding apparatus for enhanced oil recovery: system design, core-holder stress and sealing, thermal and flow analysis

MPrep Engineering - case study 04, October 2026. Chemical EOR (surfactant-polymer) and CO2-EOR laboratory rig: system design, parametric Hassler-type core holder (FreeCAD), ASME VIII-2 design-by-analysis with CalculiX, sleeve sealing, transient thermal, OpenFOAM inlet-distributor CFD and a Buckley-Leverett process model that ties the rig specification to the floods it has to run.

Design basis: typical industry values, not a client specification. Generic specifications; no vendor is endorsed. Concept design, not fabricated or tested.

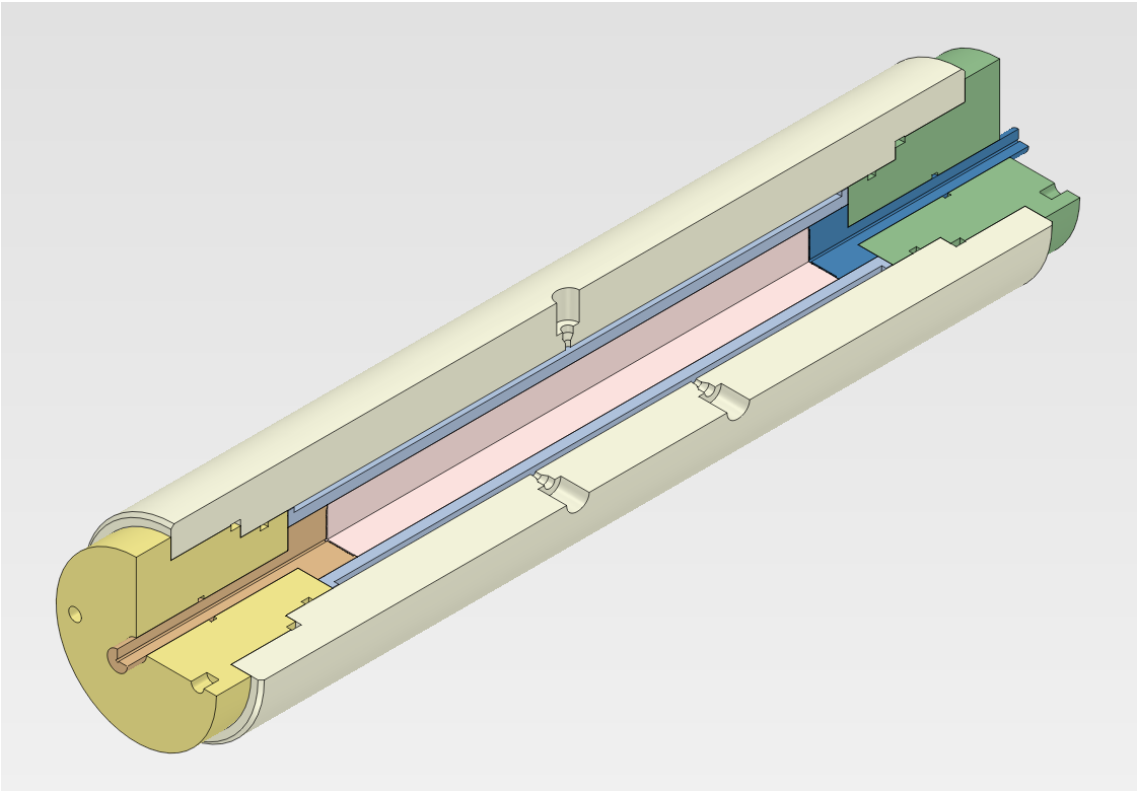


Figure 0 - Hassler-type core holder, quarter cut-away of the FreeCAD STEP model (rendered in PrePoMax): body, threaded end plugs, floating inlet end piece, sleeve, 1.5 in x 12 in core.

1. Summary

The rig injects brine, oil, HPAM polymer, surfactant and CO2 from four 1000 mL floating-piston transfer vessels, driven by a pulse-free dual-cylinder syringe pump at 0.01-20 mL/min, through a 1.5 in x 12 in core held at up to 10 000 psi confining / 9 000 psi pore pressure and 150 C (177 C design). A Darcy envelope sets four dP transducers (5 / 50 / 500 / 1500 psid) and shows why the useful dP is capped by the net confining stress, not by the pump. Relief set points follow the multiple-device rule: PSV-102 at MAWP, RD-101 at 105 % MAWP on the holder, PSV-101 at 9,500 psi on the pore side.

The 4.5 in OD body in annealed SA-479 316 passes ASME VIII-2 at 10 000 psi and 177 C (general membrane 79.6 MPa vs S = 102.7 MPa). Two findings drive the material choice. First, the confining-port cross-bore uses **153 of the 154 MPa** allowed for local primary + bending stress, i.e. no margin. Second, at the 16,640 psi hydrotest the bore yields locally and the port ligament exceeds the test limits when they are applied to the local SCL (PL 223 vs 0.95 Sy = 195 MPa). C-276 removes both issues and is the right wetted material for hot chloride brine with CO2 and traces of H2S, so it is the recommendation; 316 remains acceptable for sweet, low-chloride service. Fatigue at 20 000 pressure cycles is not limiting (usage 0.47 at the cross-bore, 0.34 at the body thread root).

The sleeve seals along the core once the net confining stress exceeds the stress needed to press the rubber into the core surface roughness: 0.54 MPa (78 psi) for 20 um / 1 mm waviness on a 75 Shore A FKM sleeve, within 9 % of the Johnson complete-contact solution. This is the physical basis of the familiar lab rule of "at least 500 psi net", which adds margin. Along the core the contact pressure averages 1.04 x the net stress and dips to 0.83 x just beyond the chamfered

joint, so the net stress needed is about 94 psi (20 um) to 280 psi (50 um); rougher or vuggy faces and the higher inlet pore pressure during a flood push the practical setting to 500-1000 psi.

Check	Criterion	Result	Verdict
Body general membrane Pm (SS316, 177 C)	<= S = 102.7 MPa	79.6 MPa	pass
Thread root / bore-step SCLs (SS316)	PL+Pb <= 1.5S = 154 MPa	max 78 MPa	pass
Confining-port cross-bore (SS316)	PL+Pb <= 154 MPa	153.4 MPa	pass, no margin
Confining-port cross-bore (C-276)	PL+Pb <= 241 MPa	153.3 MPa	pass
Hydrotest (VIII-2 8.2.1), SS316	PT = 16,640 psi; Pm <= 0.95 Sy	132.5 / 195 MPa	pass (local bore yield)
End-plug thread 3-8 UN, 12 threads	shear <= 0.6 S	SF 1.97 (plug thread governs)	pass
Fatigue, 20 000 cycles 0 -> 10 000 psi	usage <= 1.0	0.47 cross-bore / 0.34 thread root	pass
Sleeve sealing (20 / 50 um roughness)	contact pressure > closure pressure	net >= 94 / 280 psi (macro run to 381 psi)	set 500-1000 psi
Heat-up to +/-1 C, oven	-	593 min	slow, very uniform
Heat-up to +/-1 C, jacket + insulated ends	-	108 min	fast
Distributor uniformity at 1 mm into the core	UI (1 = uniform)	spiral 0.86 / rings+radial 0.75 / radial 0.66	spiral best
Port ligament at hydrotest, SS316	test limits applied to local SCL	PL 223 / 195 MPa	exceeded: use C-276

2. Design basis (typical industry values, not a client specification)

Parameter	Value	Note
Core plug	38.1 mm (1.5 in) x up to 305 mm (12 in)	1 in cores via sleeve / end-piece adapters (not analysed)
Confining pressure (MAWP)	10,000 psi (68.9 MPa)	core holder design pressure
Pore pressure, max	9,000 psi (62.1 MPa)	
Net confining stress in operation	500-1000 psi	checked in section 7
Temperature	150 C operating, 177 C (350 F) design	allowables at 177 C
Flow rate	0.01-20 mL/min	1 ft/day frontal velocity = 0.048 mL/min (phi 0.20)
Fluids	brine (high-chloride), crude oil, HPAM, surfactant, CO2 (gas / supercritical)	H2S traces: sour-service note only
Fatigue	20,000 pressure cycles	assumed: 8 cycles/day, 250 days/year, 10 years

3. System design

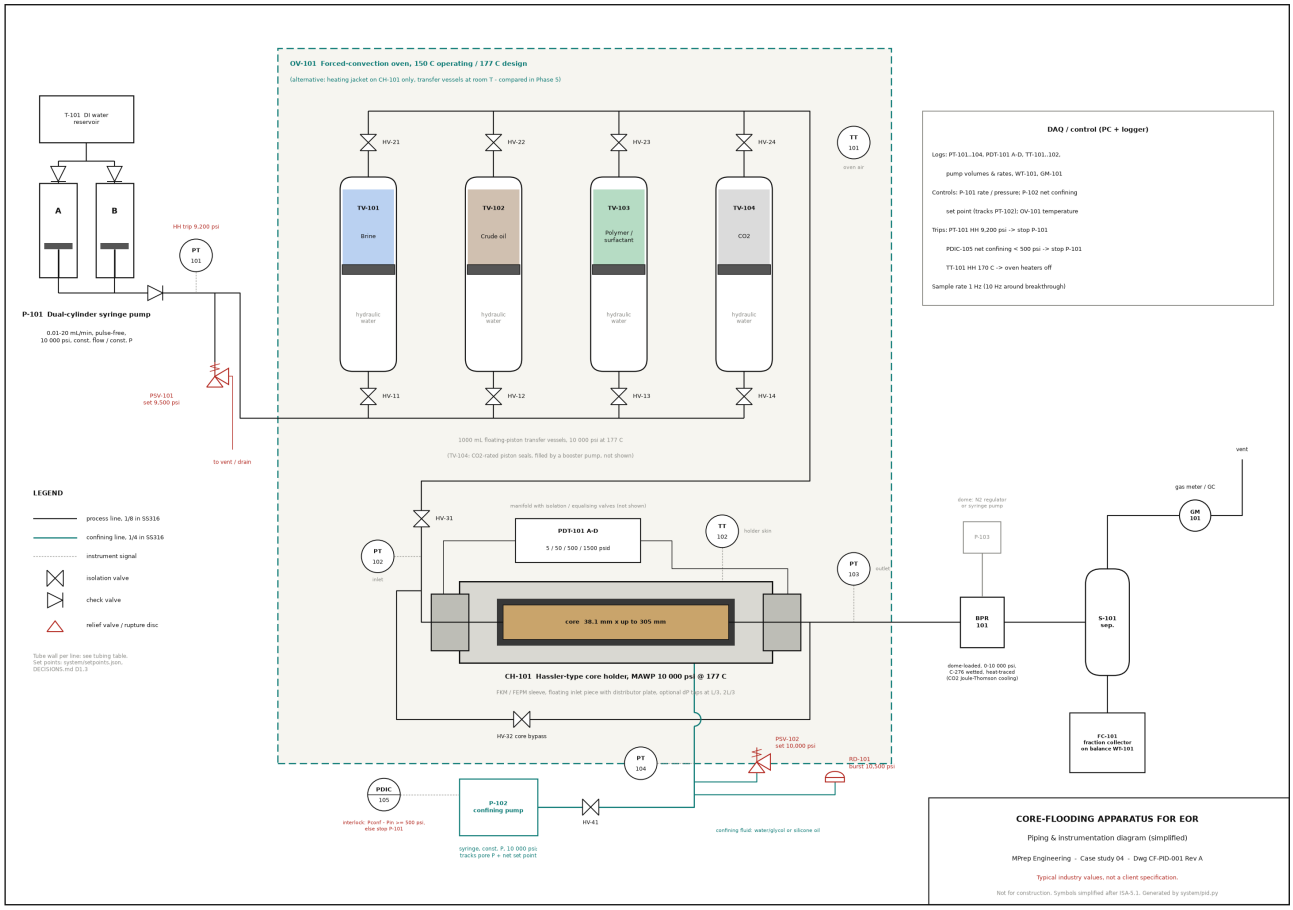


Figure 1 - Simplified P&ID; (scripted, system/pid.py). Oven OV-101 holds the four transfer vessels and the core holder; P-102 tracks the inlet pressure plus a net-confining offset; relief devices in red.

Rig architecture. The dual-cylinder syringe pump P-101 pushes de-ionised water under the floating pistons of TV-101...104, so the test fluids never touch the pump. A two-valve manifold selects the fluid; a bypass lets the lines be flushed and the BPR set without flowing through the core. The confining pump P-102 runs in constant-pressure mode and follows the inlet pressure (PT-102) plus a net-confining set point, so the sleeve load stays constant while the pore pressure changes. The dome-loaded back-pressure regulator holds the pore pressure; for CO2 it is heat-traced because CO2 expanding from 9 000 psi at 150 C cools to -29 C across it (Joule-Thomson, Span-Wagner EOS). Effluent goes to a separator, a fraction collector on a balance, and a gas meter.

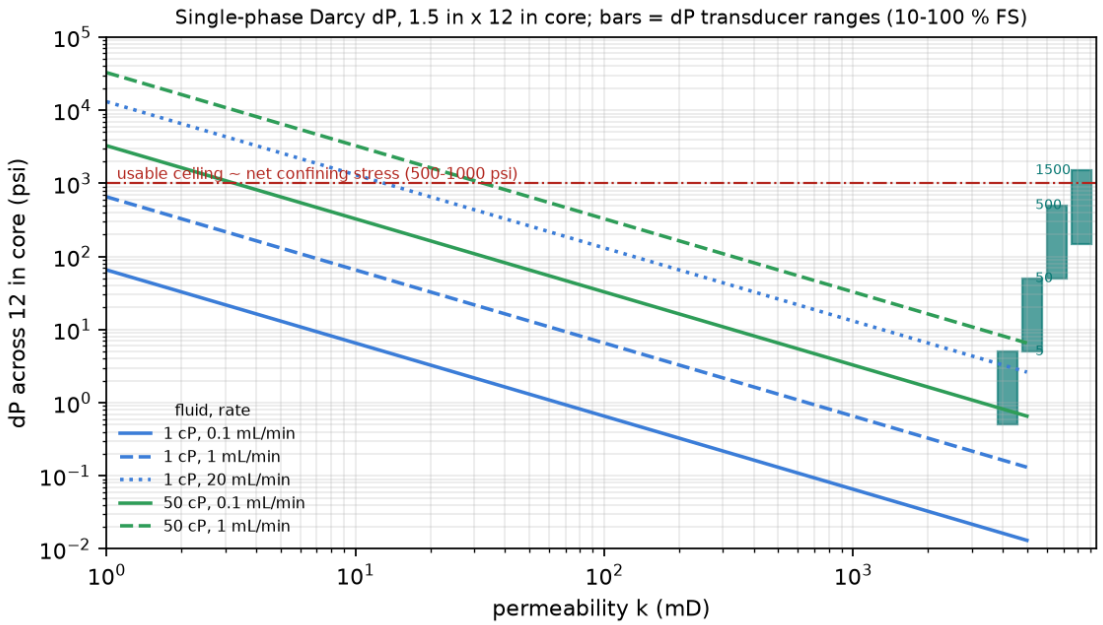


Figure 2 - Single-phase Darcy dP over the 12 in core. Transducer ranges (bars, 10-100 % FS) cover every case between 0.5 psi and 1500 psi at <= 1 % of reading. Cases above ~1000 psi cannot be run anyway: dP above the net confining stress lifts the sleeve at the inlet.

Tubing and fittings (ASME B31.3). Rated pressure from eq. (3a) with a -15 % wall tolerance (ASTM A269, OD < 1/2 in). S for TP316 at 177 C: 103.0 MPa on the 2/3 Sy basis (x0.75 of the room-temperature 137.9 MPa), 137.9 MPa on the 0.9 Sy basis that B31.3 Table A-1 permits for austenitic steel. These S values were reconstructed from the ASME II-D yield trend and must be checked against the edition in force.

Tube (OD x wall)	t/D	P, RT (psi)	P, 177 C, 2/3 Sy	P, 177 C, 0.9 Sy	P, 177 C, B31.3 Ch. IX
0.125 in x 0.028 in	0.224	8,985	6,708	8,985	6,197
0.125 in x 0.035 in	0.280	11,759	8,780	11,759	8,356
0.25 in x 0.049 in	0.196	7,689	5,741	7,689	5,240
0.25 in x 0.065 in	0.260	10,739	8,018	10,739	7,543
0.25 in x 0.083 in	0.332	14,579	10,886	14,579	10,745
0.25 in x 0.0705 in	0.282	11,863	8,858	11,863	8,440

In the hot zone the pore circuit uses 1/8 x 0.035 in (adequate for the 9 500 psi PSV set point on the 0.9 Sy basis) and the confining circuit 1/4 x 0.083 in. 1/8 x 0.028 in and 1/4 x 0.049 / 0.065 in are not adequate at 177 C and 10 000 psi. Compression fittings are rated to the tube; valves must carry a 177 C rating.

Overpressure protection and stored energy.

Device	Circuit	Set	Basis
Pump HH trip	injection	9,200 psi	below PSV-101 so it does not simmer
PSV-101	injection / pore	9,500 psi	pore <= Pconf,max - 500 psi
PSV-102	confining	10,000 psi	= MAWP (single-device rule)
RD-101	core holder confining port	10,500 psi at 177 C	second device <= 105 % MAWP; trapped-liquid expansion
PDIC-105	pore vs confining	net < 500 psi -> stop P-101	protects sleeve seal and core

A 1 L CO2 vessel at 9 000 psi stores 69 kJ at 150 C (21 kJ at 20 C) of isentropic expansion energy, about 79 times the same vessel full of water (0.9 kJ); the ideal-gas formula overstates it to 163 kJ. Shields at the oven door and BPR, minimum CO2 inventory, remote operation during CO2 runs, full bleeding of gas from the confining circuit, and depressurisation of pore before confining (and at <= 2 MPa/min after CO2 exposure, section 7) are part of the operating procedure. Trace H2S at 9 000 psi exceeds the 0.3 kPa partial-pressure threshold of NACE MR0175 / ISO 15156, which is one more reason for C-276 wetted parts.

4. Core holder design and hand calculations

The holder is a Hassler design: the confining fluid acts on the outside of an elastomer sleeve that grips the core, and the inlet end piece floats axially, so the core sees the same net stress radially and axially. Body, end plugs (3-8 UN, 12 threads), end pieces, sleeve, ports and seals are generated from the design-basis file by one FreeCAD script and exported as STEP; the 2D drawings come from section cuts of the same model.

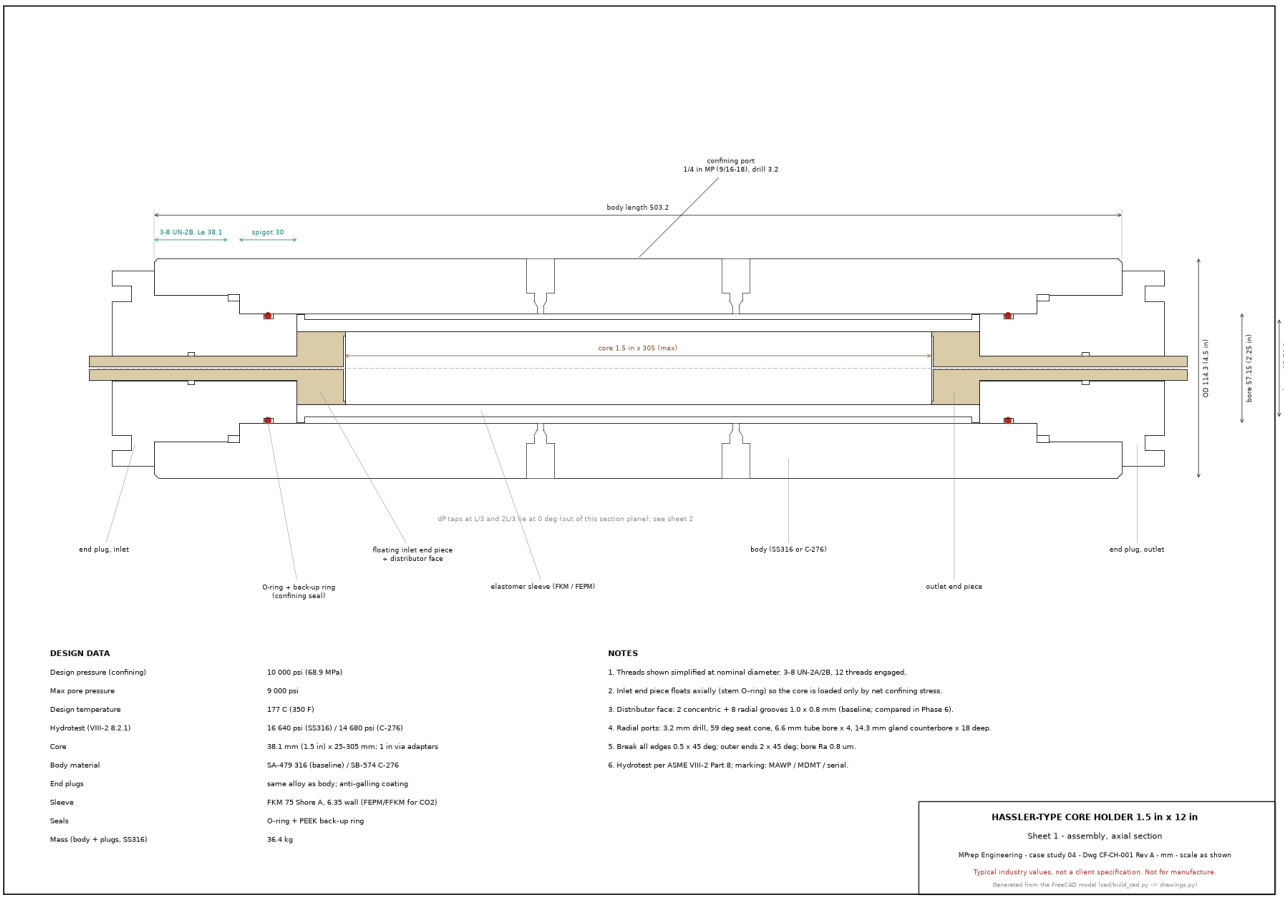


Figure 3 - General arrangement, axial section (drawing CF-CH-001, generated from the FreeCAD model).

Wall thickness (VIII-2 4.3.3) and Lame. S = min(Su/2.4, Sy/1.5) at 177 C. Elevated-temperature properties come from the ASTM minima and published temperature trends (ASME II-D trend for 316, data sheet for C-276); verify against II-D for the edition in force.

Material	S (MPa)	t req. / OD req.	Pm / S	PL+Pb / 1.5S	Hydrotest PT	Test Pm / limit	Collapse margin
SA-479 316	103	27.4 / 111.9 mm	0.78	0.70	16,640 psi	132 / 195	1.79
SB-574 C-276	161	15.3 / 87.8 mm	0.50	0.45	14,678 psi	117 / 269	2.80
Ti-6Al-4V (non-code)	310	7.1 / 71.4 mm	0.26	0.23	15,057 psi	120 / 787	7.50

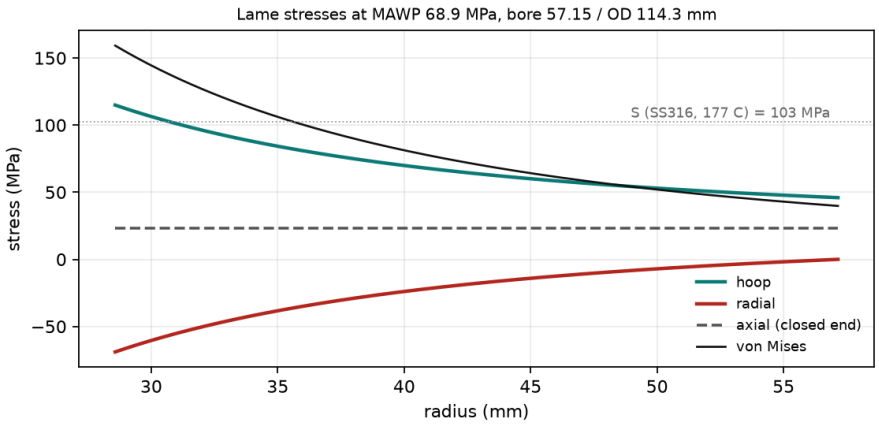


Figure 4 - Lame stresses at 10 000 psi in the 57.15 / 114.3 mm body. Bore hoop 114.9 MPa; peak von Mises at the bore 159.2 MPa, slightly above Sy(177 C) = 154 MPa: the bore of annealed 316 yields locally at 9,672 psi.

Division 2 vs Division 3. At 10 000 psi the Div 2 rules still give a sensible wall (k = 2.0). Div 3 is the better route above roughly 10-15 ksi, where the Div 2 thickness grows exponentially, or when autofrettage, fracture mechanics or a leak-before-break argument are wanted. Strictly, a laboratory core holder of this size is often outside the stamping scope; the code is used here as the engineering standard.

End-plug thread. The plug carries P on the 57.15 mm seal diameter: F = 176.9 kN. Shear areas for 3-8 UN-2A/2B over 38.1 mm (ASME B1.1 / FED-STD-H28 formulas): body (internal) 7,058 mm2, plug (external) 5,652 mm2. Shear stress 25.1 / 31.3 MPa vs 0.6 S = 61.6 MPa: safety factors 2.46 (body) and **1.97 (plug, governing)**. The minimum engagement for the body thread is 15.5 mm; 38.1 mm is kept for galling margin. FEA (section 5) shows the inner teeth carry most of the load, so the extra length mainly adds galling margin rather than strength. Same-alloy threads gall: use a silver or Ni-PTFE plated plug, or a dissimilar-hardness pair, plus nickel anti-seize.

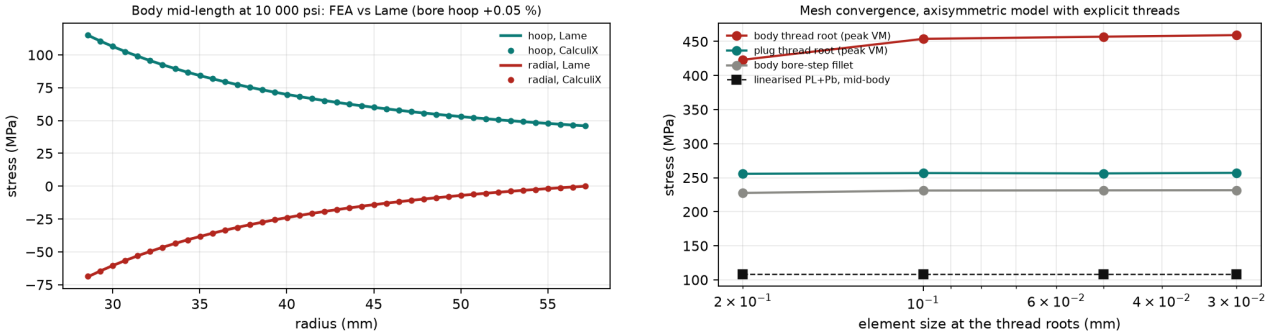
X-ray / CT-transparent variant. Transmission through two walls (Beer-Lambert, NIST XCOM mass attenuation coefficients) with the wall each material needs:

Body	Wall (mm)	T at 100 keV	T at 150 keV
Fe/316	27.4	8.5e-06 %	0.019 %
Ni/C-276	15.3	0.00056 %	0.24 %
Ti-6Al-4V	7.1	18 %	35 %
Al 7075	15.4	23 %	30 %
CFRP (C)	15.0	50 %	53 %

Steel and nickel bodies are opaque to lab CT sources. Ti-6Al-4V needs only a 7.1 mm wall and lets ~18 % through at 100 keV, but Ti Gr 5 is not listed in ASME II-D for VIII (non-code, design by the same margins and by test). Aluminium is limited to ~120 C. A carbon-fibre (filament-wound) body is the most transparent and is qualified by burst test. For CO2 or sour service, titanium also needs a check against hydrogen pick-up.

5. Stress analysis (CalculiX, PrePoMax-compatible)

Axisymmetric model. Body and inlet end plug, half length (symmetry plane at mid-body), with the UN thread teeth modelled explicitly: UNR root radius 0.144p on the plug, 0.25 mm assumed on the body, 0.1 mm clearance on the unloaded flank, frictionless node-to-surface contact on the load flanks. CAX8 elements, local refinement to 0.03 mm at the three innermost teeth and the fillets. Loads: 10 000 psi on the bore beyond the O-ring, on the spigot, and on the plug face together with the end-piece load (P on the full 57.15 mm seal area). A weak axial spring removes the plug rigid-body mode before contact closes. The axial reaction at the symmetry plane is 175.9 kN against 176.9 kN from the hand calculation.



Body mid-length: bore hoop 114.98 MPa vs Lame 114.91 MPa (+0.05 %). Thread-root peak stress vs element size: 454 / 457 / 459 MPa at 0.10 / 0.05 / 0.03 mm. Linearised stresses do not move.

Figure 5 - Verification of the axisymmetric model.

Stress classification line (SS316, 10 000 psi, 177 C)	PL (MPa)	limit	PL+Pb (MPa)	limit	peak VM
body_mid (Pm)	79.6	103	107.9	154	159
body at bore step fillet	19.3	154	67.8	154	194
body through innermost thread root	25.9	154	77.5	154	421
plug undercut (relief)	42.5	154	63.9	154	89
plug spigot root fillet	54.4	154	60.7	154	185
plug through innermost thread root	40.7	154	61.5	154	97

Linearisation follows VIII-2 Annex 5-A (membrane and bending per component, von Mises equivalent; bending of the stress normal to the SCL and of the shear stress excluded). Away from the threads the classification is general primary membrane (limit S); at the threads and fillets it is local (limit 1.5 S). Every SCL passes with a wide margin: the thread region is governed by peak stress, i.e. fatigue.

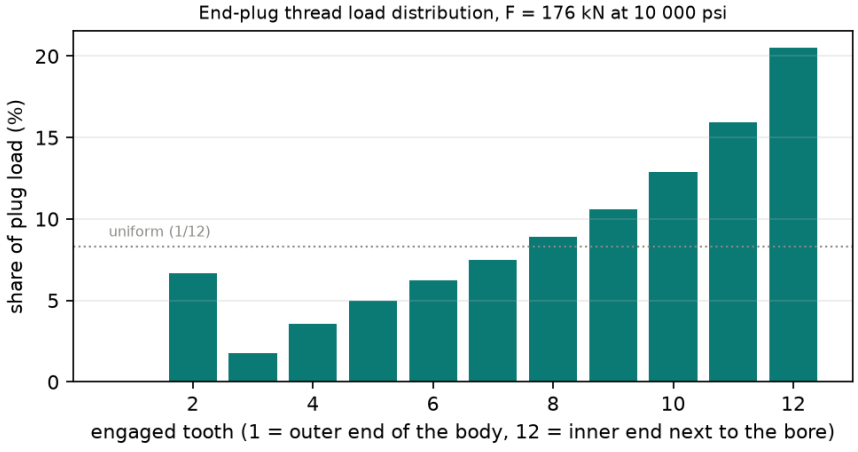


Figure 6 - Share of the end-plug load per engaged tooth. The innermost tooth carries ~21 %, about 2.5 times a uniform share; the outer three carry almost nothing.

Hydrotest. At PT = 16,640 psi (room temperature) the body membrane is 132.4 MPa against 0.95 Sy = 195 MPa, and the largest local SCL is 180 MPa against the test PL+Pb limit. The thread-root and bore peaks exceed yield locally, so the hydrotest leaves compressive residual stress there; this is beneficial and expected. The highest nodal von Mises value in the thread model (519 MPa at design pressure) sits at the sharp corner where the plug crest meets the loaded flank: a contact-edge singularity of the idealised crest, in compression and mesh-dependent, so it is excluded; the tensile thread

root (459 MPa, converged) is assessed.

C-276 body. Elastic stresses are almost the same as in 316 (stiffness differs by 6 %): body Pm 79.6 MPa vs S = 161 MPa, thread-root peak 458 MPa vs Sy(177 C) = 241 MPa.

3D model of the confining port. Quarter model of the body around the radial port (C3D10, symmetry planes through the port axis), with the realistic 1/4 in medium-pressure gland: 3.2 mm drill, 59 deg seat cone, 6.6 mm tube bore, 14.3 mm x 18 mm gland counterbore. Pressure acts on the bore, drill and seat cone; the far end carries the closed-end axial stress. The hoop stress at the bore / cross-bore corner is 346 MPa, an SCF of 3.01 on the Lame bore hoop, Hand check: a small hole in a 2:1 biaxial field has Kt = 3 - 0.5 = 2.5, and the pressure on the hole wall adds about P, so 2.5 x 115 + 69 = 356 MPa, within 3 % of the FEA. The peak von Mises stress (499 MPa) sits where the drill meets the seat cone.

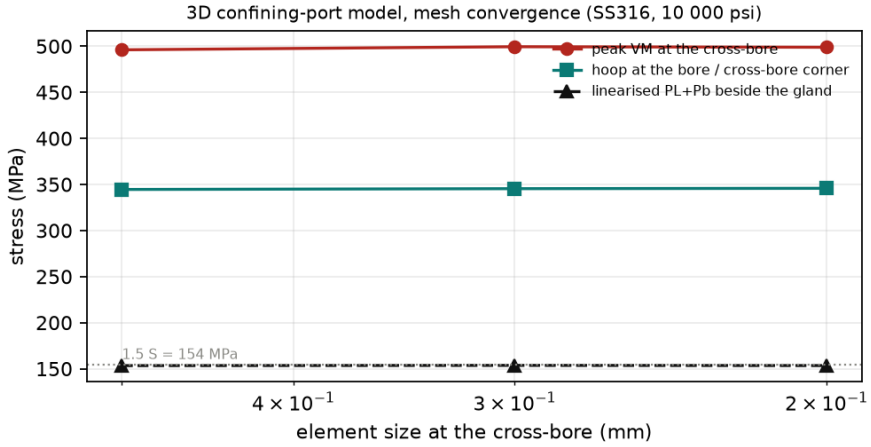


Figure 7 - Port model convergence (0.5 / 0.3 / 0.2 mm at the cross-bore). The linearised PL+Pb on the SCL beside the gland counterbore converges at 153.4 MPa, against the 154 MPa limit for 316.

The SCL through the ligament beside the gland counterbore gives PL+Pb = 153.4 MPa against 1.5 S = 154 MPa for 316: it passes, with zero margin. In C-276 the same detail gives 153.3 MPa against 241 MPa. At the 16,640 psi hydrotest the same SCL in 316 reaches PL = 223 MPa against 0.95 Sy = 195 MPa (test limits applied conservatively to the local SCL): the ligament yields during the test. Options for a 316 body: a C-276 body (recommended for the fluids anyway), a shallower gland, or a local boss. The largest value in the port model (713 MPa) is at the single pinned node at the far end, a point-constraint artefact 60 mm from the port.

6. Fatigue under pressure cycling

Each test pressurises the holder from 0 to 10 000 psi and back; 20,000 cycles are assumed (8 per day, 250 days a year, 10 years). The VIII-2 Part 5.5.3 elastic route is followed: peak equivalent stress range from the FEA with the notches modelled (Kf = 1); Ke = 1 because the linearised range is far below SPS = 3S; E corrected to the curve modulus; modified Goodman with mean-stress relaxation where the elastic peak exceeds yield. The curve is the strain-life mean curve for 304/316 stainless steel in air (Chopra & Smith 1998), ln(N) = 6.703 - 2.030 ln(ea - 0.126), with the classic ASME margins of 2 on strain and 20 on life. This is the documented basis of the code design curve, not the Annex 3-F table itself, and must be replaced by Table 3-F.10 for a code calculation.

Location	dS peak	dS lin / SPS	Salt,eq	N allowed	Usage
Body internal thread root (innermost tooth)	459	78 / 308	241	59,327	0.337
Plug external thread root (UNR)	257	62 / 308	140	2,917,602	0.007
Body fillet, thread relief -> bore step	232	68 / 308	130	> 1e7	0.001
Plug fillet, undercut -> spigot	287	61 / 308	152	1,042,687	0.019
Body bore, mid-length (Lame)	159	108 / 308	97	> 1e7	0.000
Confining port: bore / cross-bore intersection	499	153 / 308	262	42,718	0.468

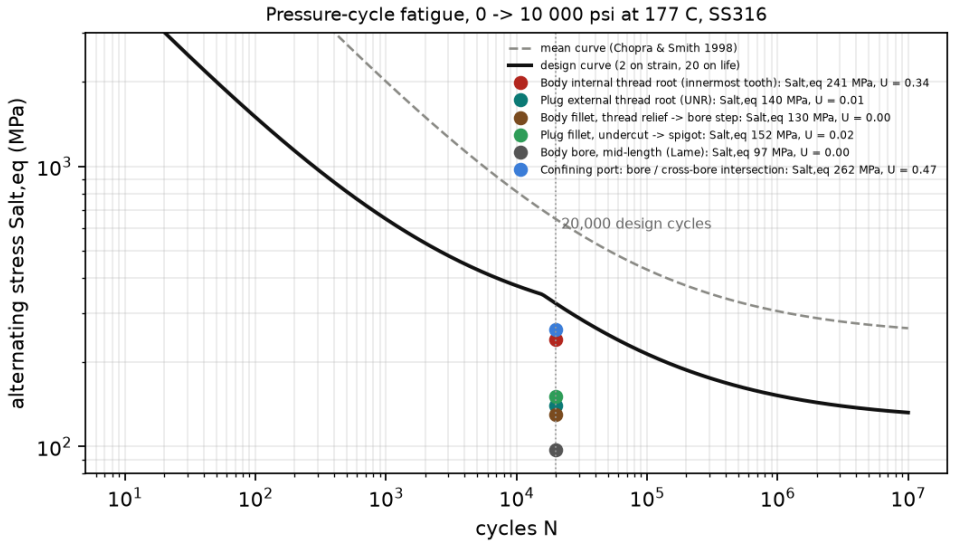


Figure 8 - Design fatigue curve and the assessed locations at 20 000 cycles (stresses in MPa).

The cross-bore governs (usage below 0.5) and the body thread root at the innermost tooth comes second. Both have margin for about 40 000 full-pressure cycles. A rig that cycles more often should use smooth, reamed cross-bore edges and a radiused internal thread root; both details are already assumed here.

7. Sleeve sealing and the minimum net confining stress

Bypass flow along the core wall happens where the sleeve does not press into the core surface hard enough to close the gaps between asperities. Two models answer the question:

Micro model (roughness). A plane-strain strip of the 6.35 mm sleeve wall (Mooney-Rivlin, C10 = 0.940, C01 = 0.235 MPa, D1 = 0.0043, from Shore A 75 via Gent's relation) is pressed onto a sinusoidal core surface. Full contact (no open channel in the valleys) is reached at 0.536 MPa for 20 um amplitude and 1 mm wavelength. Johnson's complete-contact pressure for a sinusoidal surface on an elastic half-space, p* = pi E* Delta / lambda, gives 0.591 MPa. The FEA is 9 % lower because the sleeve is a finite layer rather than a half-space. Halving the mesh size changes nothing (0.536 MPa).

Case	Amplitude / wavelength	Sleeve	Full contact (FEA)	Johnson p*
mi3	20 um / 1.0 mm	E 7.1 MPa	0.54 MPa (78 psi)	0.59 MPa
mi4	20 um / 1.0 mm	E 7.1 MPa	0.54 MPa (78 psi)	0.59 MPa
mi5	50 um / 1.0 mm	E 7.1 MPa	1.60 MPa (232 psi)	1.48 MPa
mi6	10 um / 0.5 mm	E 7.1 MPa	0.54 MPa (78 psi)	0.59 MPa
mi7	20 um / 1.0 mm	E 20.8 MPa	1.67 MPa (243 psi)	1.75 MPa

Macro model (along the core). Sleeve, core and floating inlet end piece, axisymmetric, with the chamfered joint between core and end piece (0.5 mm and 0.3 mm). The run converged up to 2.6 MPa (381 psi) net and then stopped on the increment-size limit; the saved increments are used. Contact pressure is taken from the sleeve radial stress at its inner surface (sigma_r = -p at a frictionless interface), because node-to-surface contact pressures on quadratic elements alternate between corner and mid-side nodes. Along the core it averages 1.04 x the net stress and dips to 0.83 x about 1 mm beyond the chamfer (nodal values oscillate about +/-8 %). At the joint the sleeve bridges the chamfer groove, but that groove runs round the circumference and opens no axial path, so it does not cause bypass.

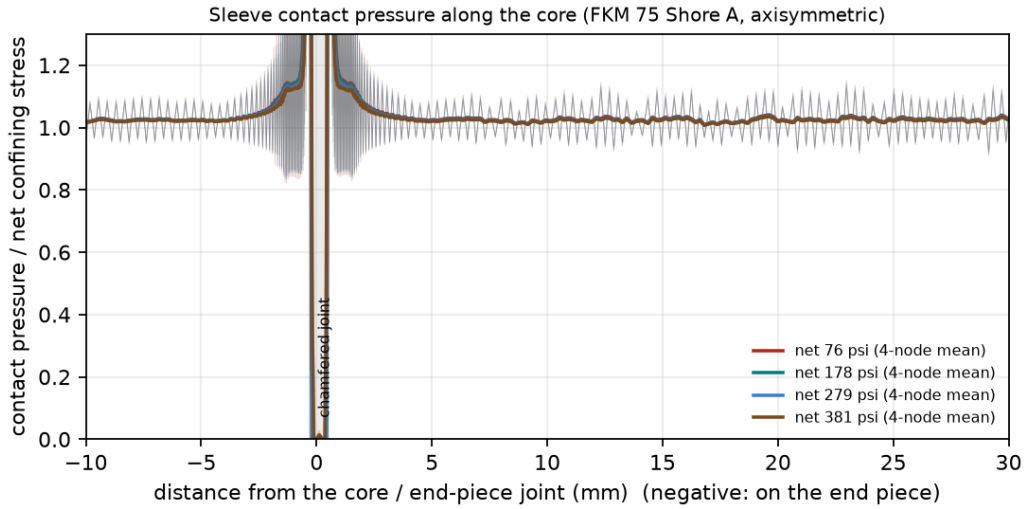


Figure 9 - Sleeve contact pressure along the core at several net confining stresses (macro model; light lines raw nodal values, bold 4-node mean).

Combining both models, net >= closure pressure / 0.83: 20 um / 1.0 mm: 94 psi; 50 um / 1.0 mm: 280 psi; 10 um / 0.5 mm: 94 psi.

Minimum net confining stress. The requirement is "contact pressure everywhere above the roughness closure pressure", evaluated at the *inlet*, where the pore pressure is highest during a flood: net(inlet) = Pconf - Pin = Pconf - Pout - dP. For saw-cut or ground sandstone faces (about 20 um) about 80 psi would be enough in theory. Rougher surfaces (50 um), harder sleeves (FFKM ~ 90 Shore A) and some margin for local defects make **500 psi at the inlet** a sound minimum, and 1000 psi for vuggy carbonates. This matches common laboratory practice and is the reason the confining pump tracks the inlet pressure instead of holding a fixed value.

Elastomer selection and rapid gas decompression. FKM (Viton) 70-80 Shore A is the baseline for brine, oil, polymer and neutral surfactant up to 150 C. FEPM (Aflas) resists alkaline-surfactant-polymer floods (pH 10-12), amines and steam, but is stiff below about 5 C, so install it warm. FFKM covers everything at many times the price. With CO2 the sleeve absorbs gas at pressure. ISO 23936-2 / NORSOK M-710 RGD qualification decompresses at ~2 MPa/min (100 C, 15 MPa, 90/10 CH4/CO2). The 6.35 mm sleeve wall is thicker than the 5.33 mm test O-ring, and desorption time grows with thickness squared, so the procedure limits depressurisation after CO2 to <= 2 MPa/min, with a hold near 7 MPa, and requires inspection of the sleeve after every CO2 test.

8. Thermal: oven vs heating jacket

Transient axisymmetric heat conduction in CalculiX (half model): steel body, plugs and end pieces, FKM sleeve, water in the confining annulus (conduction only; the 3.2 mm gap is too thin for convection to matter), brine-saturated sandstone core. Set point 150 C, ramped over 30 min. Oven: forced convection h = 15 W/m2K on all outer surfaces. Jacket: a band heater on the body OD held at the set point; the ends lose heat by natural convection + radiation (h = 10 W/m2K), or through 25 mm mineral-wool caps (h_eff ~ 1.6 W/m2K). Mesh sizes of 3 / 1.5 / 0.8 mm give the same heat-up times to within a minute.

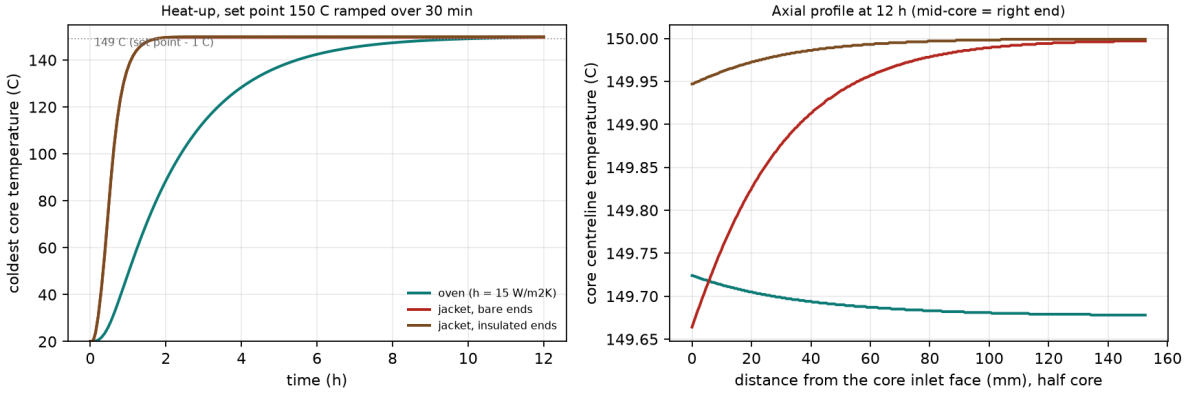


Figure 10 - Heat-up of the core (coldest point) and axial temperature spread for the three heating options.

Option	Core within +/-2 C	within +/-1 C	Final core spread	Heater power at steady state	Max dT through the wall
Oven	513 min	593 min	0.05 C	- (oven)	2.1 C
Jacket, bare ends	93 min	108 min	0.33 C	26 W	9.5 C
Jacket, insulated ends	93 min	108 min	0.05 C	4 W	9.5 C

The oven heats everything (holder, transfer vessels and lines) and leaves the core uniform to a fraction of a degree, but the 40 kg holder needs hours to settle, because h is only ~15 W/m2K over a large thermal mass. The jacket brings the core to temperature in a fraction of that time. With bare ends, the plugs lose heat and leave the core ends cooler than the middle; insulating the end caps fixes most of this. In jacket mode the injected fluid must be pre-heated: 20 mL/min of water from 20 C needs ~180 W, which a bare core end cannot supply. The radial temperature difference through the body wall peaks at 9.5 C during jacket heat-up (oven: 2.1 C). With E alpha dT / 2(1 - nu) = 2.2 MPa per C that is about 21 MPa of secondary stress, small next to the pressure stress and reached before the holder is pressurised; no separate thermal-stress FEA was needed.

9. Inlet distributor (OpenFOAM)

The floating end piece feeds the core through grooves on its face; the lands between the grooves sit on the rock, so fluid enters only under the grooves and has to spread inside the core. OpenFOAM v2512 (porousSimpleFoam, laminar), with the core as a Darcy porous zone (k = 100 mD, 15 mm modelled), the grooves 1.0 x 0.8 mm, inlet 1.6 mm, 20 mL/min of water. Mesh: structured 0.5 mm cells in plane, graded to ~0.1 mm at the core face (about 185 000 cells). A 0.25 mm mesh (730 000 cells) run for two of the patterns changes the UI by less than 0.003. The uniformity index UI = 1 - sum|u - u_mean| A / (2 u_mean A) of the axial velocity is taken at several depths.

Pattern	Groove area	UI at 0.25 mm	UI at 1 mm	UI at 2 mm	UI at 5 mm	dP vs 1D Darcy
Centre hole only	1.1%	0.108	0.277	0.397	0.672	+845 %
8 radial grooves	12.1%	0.373	0.660	0.786	0.926	+44 %
2-start spiral	27.4%	0.540	0.863	0.918	0.968	+14 %
2 rings + 8 radial (baseline)	21.5%	0.470	0.755	0.848	0.944	+21 %

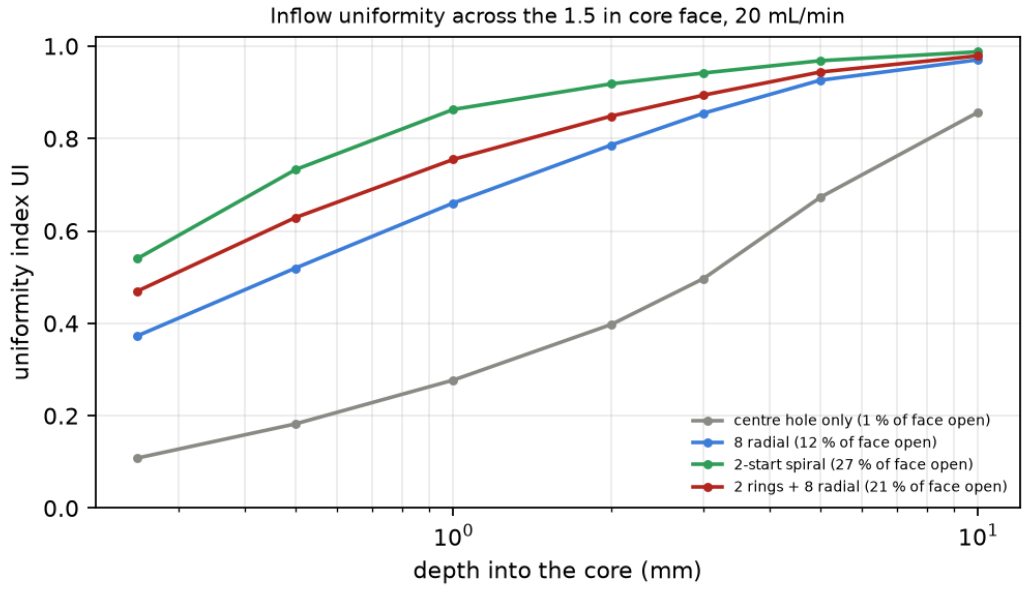


Figure 11 - Uniformity index of the axial velocity vs depth into the core for the four face patterns.

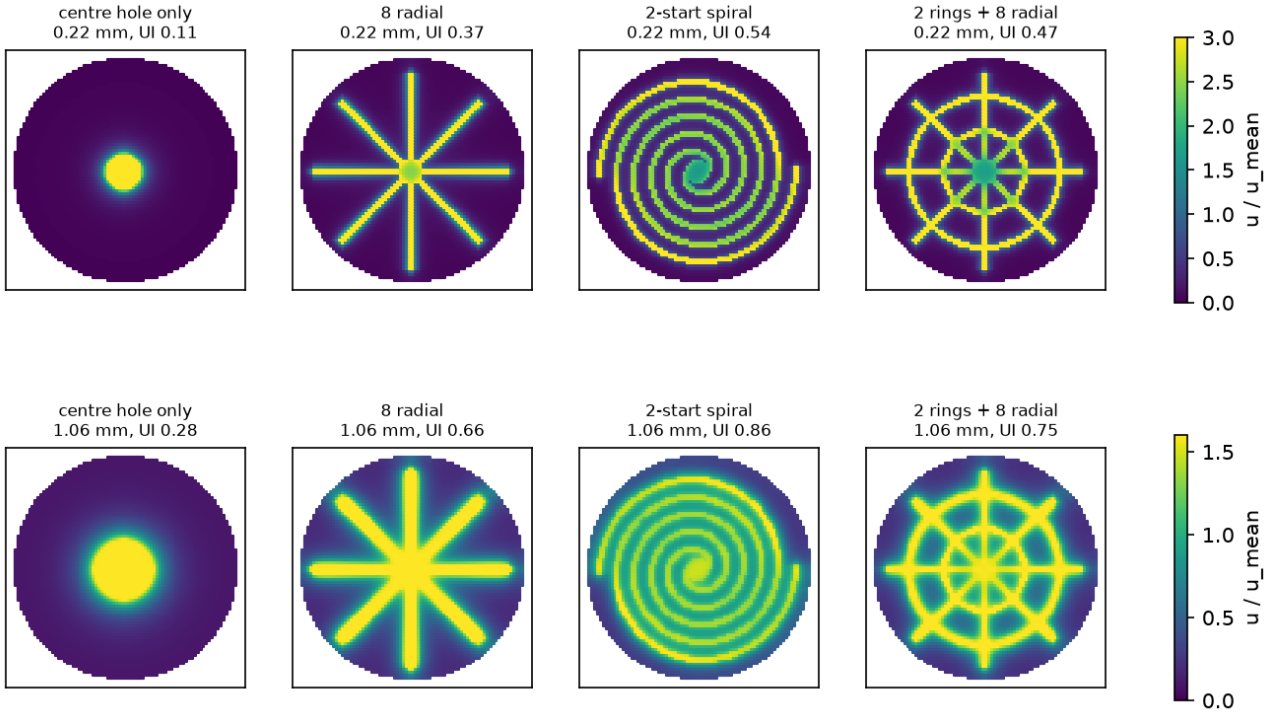
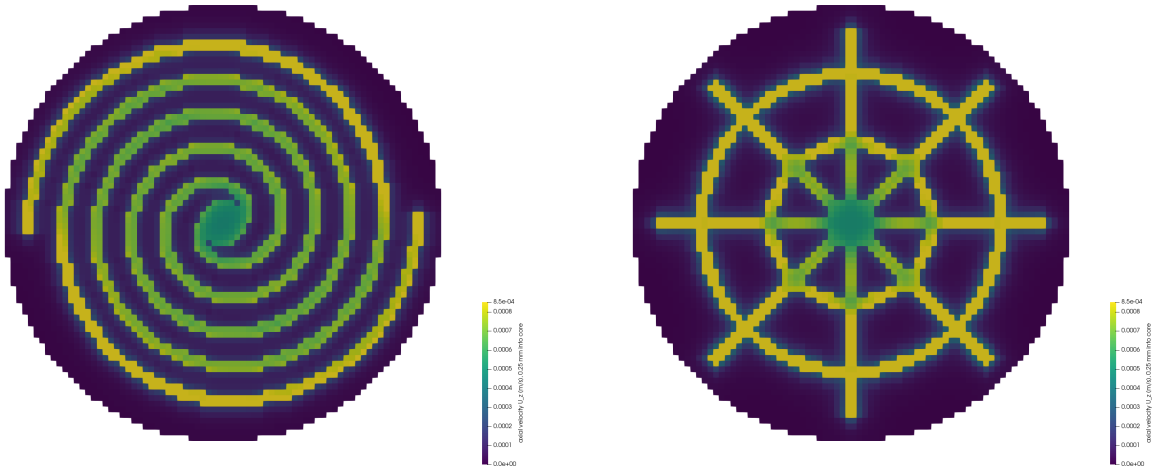


Figure 12 - Axial velocity (relative to the mean) at 0.25 mm and 1 mm into the core.



2-start spiral

2 rings + 8 radial

Figure 12b - Axial velocity 0.25 mm into the core, ParaView 5.12 (OpenFOAM results).

In a 100 mD core, groove pressure drop is negligible next to the rock, so the pattern matters through two things only: how much of the face it opens, and how far any point of the face is from the nearest groove. The 2-start spiral (6 mm pitch) opens the most area (27 %) and leaves no long unfed sectors, so it comes out best (0.86 at 1 mm); the CAD baseline (2 rings + 8 radial) is second. Part of the spiral advantage is its larger open area, so a final choice should compare patterns at equal coverage. A face with only a centre hole is unacceptable: it adds the equivalent of ~12 cm of core to the inlet dP. The extra pressure drop over plain Darcy (spreading resistance), expressed as an equivalent core length, is the error a face-to-face dP measurement carries; the dP taps at L/3 and 2L/3 avoid it. Recommendation: change the distributor face to the 2-start spiral (not yet updated in the CAD).

10. Process simulation: why the rig needs its specification

1D Buckley-Leverett fractional-flow model, Corey relative permeabilities for a water-wet sandstone (Swc 0.20, Sor 0.30, krw0 0.30, kro0 0.90, nw 3, no 2; typical values, not a measured data set), oil 10 cP. The Welge tangent solution is checked against an independent upwind finite-volume solver (they agree to 0.1 % in recovery at 3 PV). The surfactant-polymer case is qualitative: ultra-low IFT is represented only by a lower residual oil (0.08) and straighter curves.

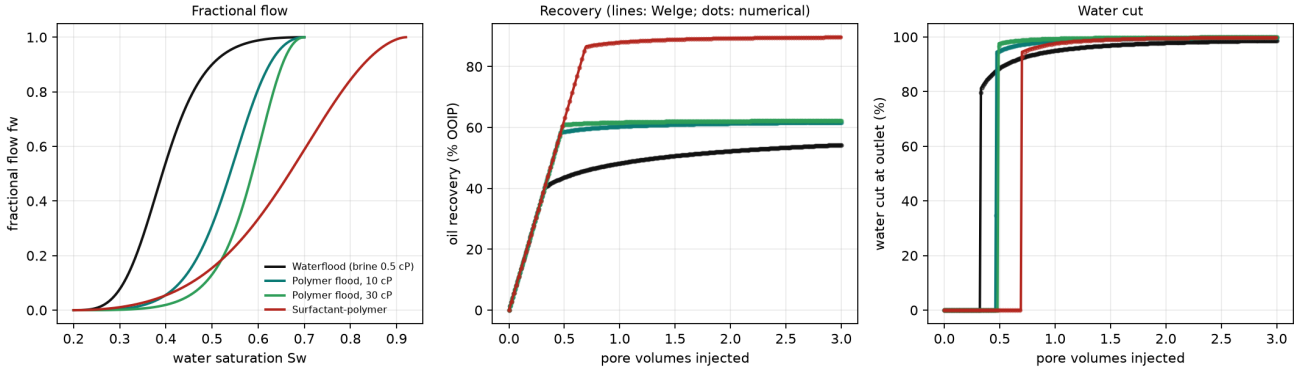


Figure 13 - Fractional flow, oil recovery and water cut vs pore volumes injected (lines: Welge; dots: numerical).

Flood	End-point M	Breakthrough (PV)	Recovery at BT	Recovery at 3 PV	Ultimate
Waterflood (brine 0.5 cP)	6.67	0.32	40 %	54 %	62 %
Polymer flood, 10 cP	0.33	0.47	58 %	62 %	62 %
Polymer flood, 30 cP	0.11	0.49	61 %	62 %	62 %
Surfactant-polymer	0.44	0.69	86 %	90 %	90 %

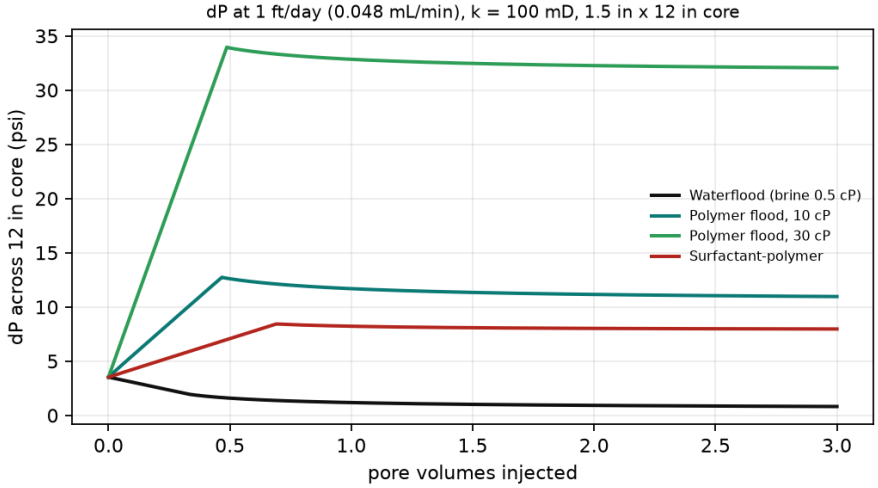


Figure 14 - Pressure drop over the 12 in core at 1 ft/day and 100 mD during each flood.

What this means for the rig: **(1) dP range.** The polymer flood raises dP several-fold at the same rate, and dP scales with q/k. A 10 mD core at 10 ft/day with 30 cP polymer reaches 3,396 psi, far beyond the net confining stress. The 5-1500 psid transducer set, and the rate limit that follows from it, come straight from this. **(2) Pressure.** The pore pressure has to reach reservoir pressure, and for CO2 it has to sit above the minimum miscibility pressure, plus a 500-1000 psi net confining stress on top: hence 9 000 / 10 000 psi. **(3) Temperature.** Reservoirs of interest for chemical EOR run at 60-120 C, and the HPAM and surfactant stability tests that decide a flood are run at or above reservoir temperature: hence 150 C, with a 177 C design margin. **(4) Low rates.** 1 ft/day in a 1.5 in core is 0.05 mL/min, which only a syringe pump delivers pulse-free.

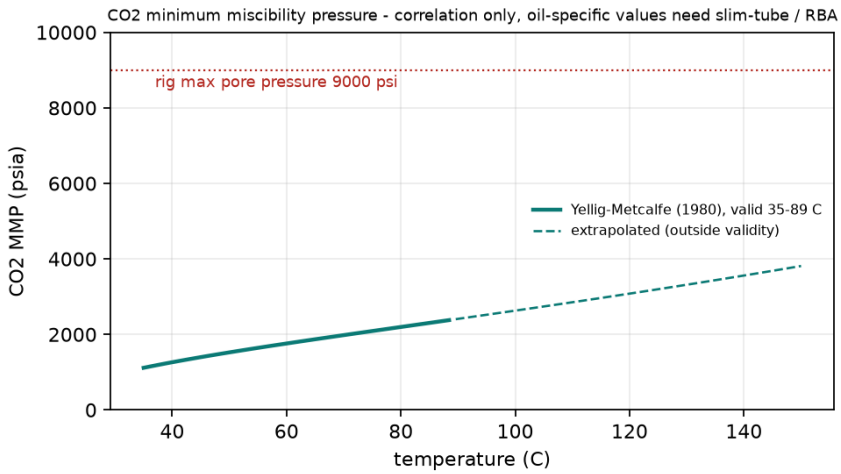


Figure 15 - CO2 MMP from the Yellig-Metcalfe correlation (valid 35-89 C, dashed beyond).

CO2-EOR note. CO2 becomes miscible with oil above the minimum miscibility pressure (MMP), when multiple-contact extraction of the light and intermediate components removes the interface. The Yellig-Metcalfe correlation gives 1,759 psi at 60 C and 2,197 psi at 80 C; extrapolated to 150 C it suggests ~3,812 psi, but that is outside its validity range and ignores oil composition. Oil-specific MMP comes from slim-tube or rising-bubble tests, not from a core flood. In the core flood CO2 viscosity is 0.03-0.1 cP, so the mobility ratio is very unfavourable: gravity override and fingering in a horizontal core are real, and vertical, top-down injection is often used. No recovery figure for CO2 is claimed here.

11. Model and results in the native tools

The views below are taken from the tools used for the study through their own scripting routes: PrePoMax 3.0 (--script, opening the CalculiX .frd files) and ParaView 5.12 pvbatch for the OpenFOAM cases (Figure 12b). The CAD view in Figure 0 is the FreeCAD STEP export rendered in PrePoMax.

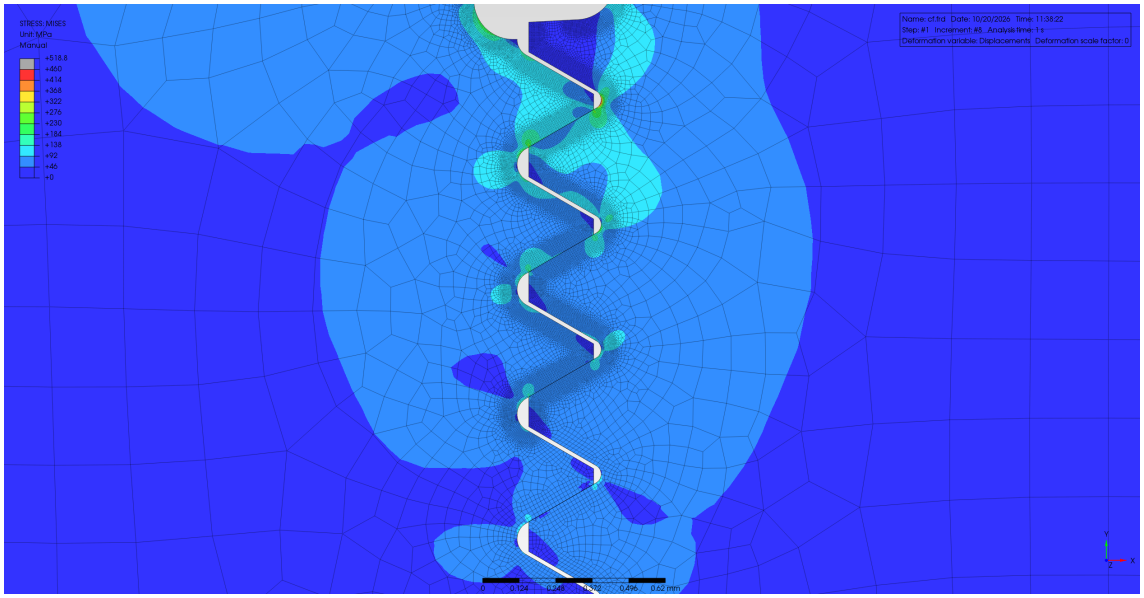


Figure 16 - PrePoMax: von Mises stress (MPa) at the end-plug threads, axisymmetric model at 10 000 psi, fixed 0-460 MPa legend (grey = above 460: the crest-corner contact singularity).

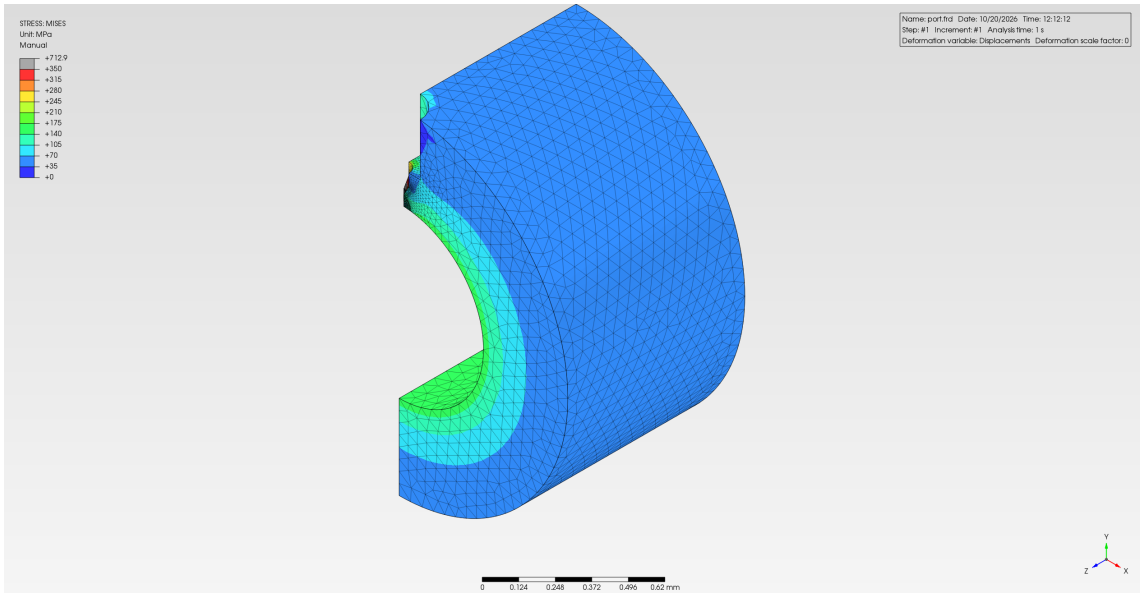


Figure 17 - PrePoMax: von Mises stress (MPa) in the 3D quarter model of the confining port, 0-350 MPa legend; the cross-bore is at the top left of the symmetry plane.

12. Verification summary

Check	Result
Lame vs axisymmetric FEA, bore hoop at mid-body	+0.05 %; OD hoop 45.96 vs 45.97 MPa
Axial equilibrium, axisymmetric model	175.9 kN reaction vs 176.9 kN plug load
Thread-root peak, mesh 0.10 / 0.05 / 0.03 mm	454 / 457 / 459 MPa
Port model, mesh 0.5 / 0.3 / 0.2 mm (PL+Pb)	153.3 / 153.6 / 153.4 MPa
Port far-field bore hoop vs Lame	117.4 vs 114.9 MPa (quadratic tets, 4 mm far field)
Sleeve micro model vs Johnson	0.536 vs 0.591 MPa; mesh x2 finer: 0.536 MPa
CFD mass balance / dP vs Darcy	inlet = outlet flow to 1e-6; dP = Darcy + spreading resistance (+21 %)
CFD mesh 0.5 vs 0.25 mm (none, radial patterns)	UI at 1 mm within 0.003
Buckley-Leverett: Welge vs numerical	recovery at 3 PV equal to 0.1 %
Thermal mesh 3 / 1.5 / 0.8 mm	heat-up times within ~1 min

13. Limitations and next steps

Allowable stresses and tube S values were reconstructed from the ASTM minima and published temperature trends; check them against ASME II-D / B31.3 Table A-1 of the edition in force before any build. The fatigue curve is the strain-life basis of the code curve, not Table 3-F.10. The sleeve uses Mooney-Rivlin constants derived from hardness (Gent); measured uniaxial + planar data for the actual FKM compound at 150 C would sharpen the sealing numbers. Contact is frictionless; threads are axisymmetric (no helix). The dP taps through the sleeve, the 1 in adapters, the end-piece stem seal and the transfer-vessel design are not analysed. Relative permeabilities are generic; a real study would use measured curves. Next steps for a project: a HAZOP and a pressure-systems review, a C-276 body drawing set, an FKM / FEPM RGD test at the actual CO2 conditions, and commissioning against a reference core with known permeability.

References

[ASME-VIII-2] ASME Boiler and Pressure Vessel Code, Section VIII, Division 2: Alternative Rules (2023). Part 4.3 (design-by-rule cylinders), Part 5 (design by analysis: 5.2.2 elastic stress analysis, 5.5 fatigue), Part 8.2 (hydrostatic test), Part 9 (overpressure protection).; [ASME-VIII-3] ASME BPVC Section VIII, Division 3: Alternative Rules for Construction of High Pressure Vessels (2023).; [ASME-VIII-1] ASME BPVC Section VIII, Division 1 (2023), UG-125...UG-140 (overpressure protection, multiple-device rule UG-134).; [ASME-IID] ASME BPVC Section II, Part D: Properties (2023), Tables 5A (Div 2 allowable), Y-1 (yield), U (tensile), TM (modulus), TE (expansion).; [B31.3] ASME B31.3-2022 Process Piping: 304.1.2 (straight pipe, eq. 3a), Table A-1 (allowable stress), Chapter IX (high-pressure piping, K304.1.2).; [A269] ASTM A269/A269M: Seamless and Welded Austenitic Stainless Steel Tubing for General Service (wall tolerance +/-15 % for OD < 1/2 in).; [A479] ASTM A479/A479M: Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels (316: Sy >= 205 MPa, Su >= 515 MPa).; [B574] ASTM B574: Low-Carbon Nickel-Chromium-Molybdenum Alloy Rod (N10276: Sy >= 283 MPa, Su >= 690 MPa).; [B348] ASTM B348/B348M: Titanium and Titanium Alloy Bars and Billets (Grade 5: Sy >= 828 MPa, Su >= 895 MPa).; [ISO15156] ANSI/NACE MR0175/ISO 15156-1/-2/-3: Materials for use in H2S-containing environments in oil and gas production.; [ISO23936] ISO 23936-2:2011 Non-metallic materials in contact with media related to oil and gas production, Part 2: Elastomers (Annex B, RGD test).; [NORSOK-M710] NORSOK M-710 Rev. 3 (2014): Qualification of non-metallic materials and manufacturers - Polymers.; [ISA-5.1] ANSI/ISA-5.1-2022 Instrumentation Symbols and Identification.; [API-RP63] API RP 63: Recommended Practices for Evaluation of Polymers Used in Enhanced Oil Recovery Operations (1990).; [API-RP40] API RP 40: Recommended Practices for Core Analysis, 2nd ed. (1998).; [ASME-B1.1] ASME B1.1-2019 Unified Inch Screw Threads (UN and UNR Thread Form).; [SpanWagner] R. Span and W. Wagner, "A new equation of state for carbon dioxide covering the fluid region from the triple-point temperature to 1100 K at pressures up to 800 MPa," J. Phys. Chem. Ref. Data 25 (1996) 1509-1596. https://doi.org/10.1063/1.555991; [CoolProp] I. H. Bell, J. Wronski, S. Quoilin, V. Lemort, "Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp," Ind. Eng. Chem. Res. 53 (2014) 2498-2508. https://doi.org/10.1021/ie4033999; [BL1942] S. E. Buckley and M. C. Leverett, "Mechanism of fluid displacement in sands," Trans. AIME 146 (1942) 107-116. https://doi.org/10.2118/942107-G; [Lake] L. W. Lake, R. T. Johns, W. R. Rossen, G. A. Pope, Fundamentals of Enhanced Oil Recovery, 2nd ed., SPE, 2014.; [Sheng] J. J. Sheng, Modern Chemical Enhanced Oil Recovery: Theory and Practice, Gulf Professional Publishing, 2011.; [Gent1958] A. N. Gent, "On the relation between indentation hardness and Young's modulus," Rubber Chem. Technol. 31 (1958) 896-906. https://doi.org/10.5254/1.3542351; [Hassler1944] G. L. Hassler, "Method and apparatus for permeability measurements," US Patent 2,345,935 (1944).; [Pipeline/Yellig] W. F. Yellig and R. S. Metcalfe, "Determination and prediction of CO2 minimum miscibility pressures," J. Pet. Technol. 32 (1980) 160-168. https://doi.org/10.2118/7477-PA; [Holm1974] L. W. Holm and V. A. Josendal, "Mechanisms of oil displacement by carbon dioxide," J. Pet. Technol. 26 (1974) 1427-1438. https://doi.org/10.2118/4736-PA; [UPMet-C276] Upper Midwest Metals (UPMet), Alloy C276 data sheet: 0.2 % yield 60 ksi at 70 F, 50 ksi at 400 F; tensile 103 ksi at 400 F. https://www.upmet.com/sites/default/files/products/datasheet/c276-datasheet.pdf; [Swagelok-tube] Swagelok, Tubing Data, MS-01-107: A269 316 tubing ratings based on S = 20 000 psi (-20 to 100 F), B31.3 method; 1/8 x 0.028 in = 8 500 psig. Used only as a cross-check of the eq. (3a) calculation.; [JamesWalker-RGD] James Walker, "RGD testing" (2022): ISO 23936-2 / NORSOK M-710 Annex B decompression rate 20 bar/min, 100 C, 150 bar, 90/10 CH4/CO2. https://www.jameswalker.biz/knowledge/insights/rgd-testing; [CCX] G. Dhondt, CalculiX CrunchiX User's Manual, version 2.22. http://www.dhondt.de/; [OpenFOAM] OpenFOAM v2512, ESI-OpenCFD, https://www.openfoam.com (porousSimpleFoam, Darcy-Forchheimer porosity).; [FreeCAD] FreeCAD 1.1, https://www.freecad.org; [gmsh] C. Geuzaine and J.-F. Remacle, "Gmsh: a 3-D finite element mesh generator," Int. J. Numer. Methods Eng. 79 (2009) 1309-1331. https://doi.org/10.1002/nme.2579